

As you have studied in the previous classes, electricity is a form of energy that can be generated when electrons or electric charges flow through a conductor like a copper wire. Electrical energy can be natural as lightning in sky or can be produced by a generator. Do you know that electricity can be used to make magnets? In the same way the opposite operation can be performed, that is magnets can be used to produce electricity.

In this Chapter you will learn about:

- Generating Electricity (Model Generator)
- Portable Generator (Bicycle Dynamo)
- Problem of Generating Electricity
- Working of Power Stations
- Other Sources of Electricity.
- Introduction to Electronic Systems
- Uses of Components (Input, Processor, Output)

All the students will be able to:

- Design an experiment to generate electricity.
- Explain the working of the model generator.
- Identify the simple devices that generate electricity in daily life.
- Design and demonstrate the working of a power station.
- List type of energy being used in power stations.
- Relate problems involved in generating electricity.
- Describe basic component of an electronic system.
- Explain A.C and D.C current.
- List component that would be needed to turn A.C to D.C.
- State how output component in various devices could be used in their schools and surroundings.

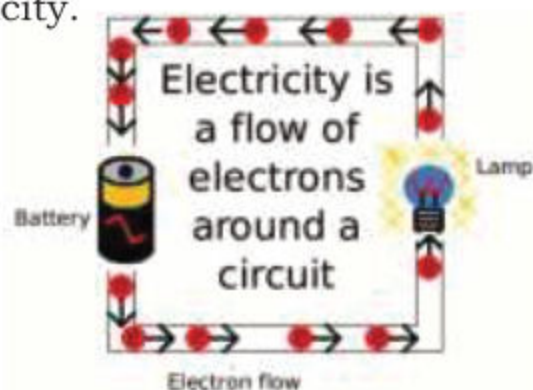


Fig.1.1: Electric Current

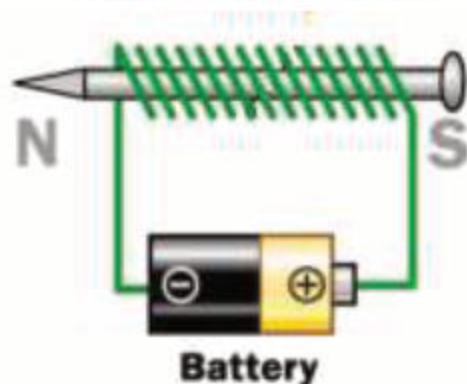


Fig.11.2: Simple Electromagnet



Fig.11.3: Power Station

GENERATING ELECTRICITY

- ✓ Design an experiment to generate electricity.

In our homes all electrical appliances use electricity and without it they cannot work. There are many methods and sources which can be used to generate electricity. For example, dry cells and batteries produce electricity by chemical reaction of compounds.

Activity 11.1:

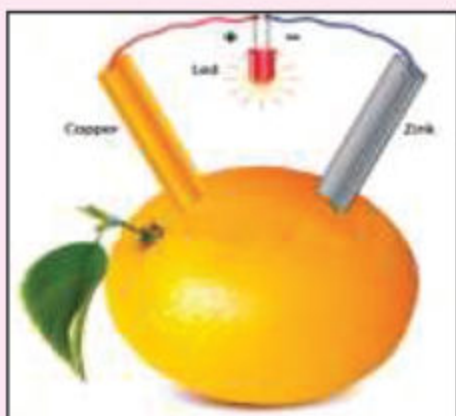
Electricity Generating Model

What I need

Copper metal piece, zinc metal piece, 1 lemon, LED.

What I do

Take two pieces of copper and zinc metals, insert the one end of both metals pieces into lemon at two different places [as shown in figure]. Connect the other end of both zinc and copper with two wires to the terminals of an LED.



Lemon battery

What I observed

I observed that LED is lighting up. The lemon produces very small current of about one milli-ampere. This current is however, very small to light up a resistance bulb.

Activity Questions

How LED lights up?

How much current is produced by the lemon?

Teacher Note:

Divide the class into two groups. Ask two students from each group to perform the activity. Rest of the students will observe the activity and will take notes. Provide one set to each pair. Ask students to perform activity shown in figure. Also perform this activity with potato and vinegar. Promote discussion on phenomenon.

ELECTRICITY GENERATOR MODEL

- ✓ Explain the working of the model generator.

We have studied above that electricity can be produced by chemical methods. Similarly, we can also generate electricity through mechanical ways, for example, moving magnet through a coil or a coil through a magnet.

For this we need to have a “U” shaped magnet and a wire with some loops called a coil. When magnet moves around the coil, the magnetic field passes through the coil and as a result, some

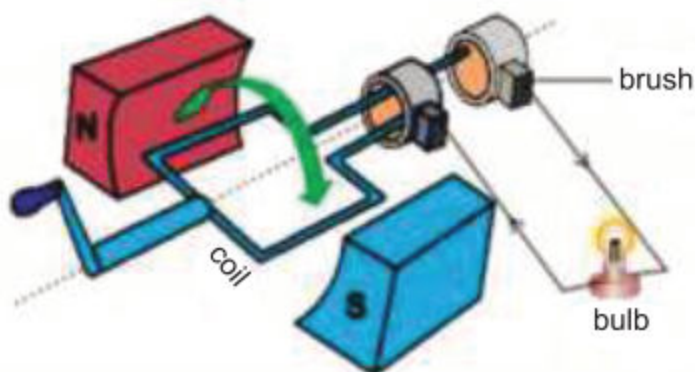


Fig.11.4: Electricity Generator Model

current passes in the coil. Both ends of the coil should be attached with some electrical appliance like a bulb, LED or a battery. As the current in coil changes its direction with an equal interval of time and therefore, this current is called Alternating Current (AC).

PORTABLE GENERATOR (BICYCLE DYNAMO)

- ✓ Identify the simple devices that generate electricity in daily life

Do you have a bicycle? If yes, then you may light a lamp to mount on it without any battery source. Do you want to know how? Let's learn.

As we know that the moving magnet through the coil can produce some amount of energy to be used for electrical appliances; here we use the same. The small generator called the bicycle dynamo is the device that can produce electric current and light up the lamp that is mounted on it.



Fig.11.5 Bicycle Dynamo Generator

The magnet inside the dynamo spins by the tyre of bicycle. As the tyre starts to rotate, magnet spins through coil, then coil starts to generate some amount of current through it. The coil end that passes attached to the lamp mounted on the bicycle. Harder you paddle the bicycle, the faster the dynamo spins. As faster it spins more electricity the dynamo produces hence bulb becomes brighter.

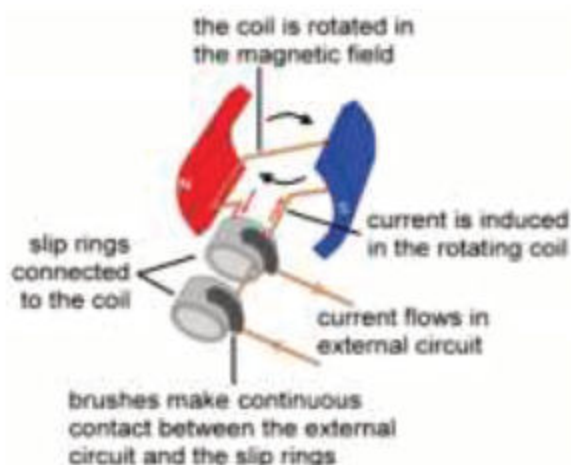


Fig.11.6: Internal View of Bicycle Dynamo Generator

WORKING OF POWER STATION

- ✓ Design and demonstrate the working of a power station.

A country requires more electricity for commercial as well as domestic use. We use power station generators for this purpose which are bigger than bicycle dynamo generator and produce electricity in a large amount to be used in homes and offices. As the dynamo has few turns of coil in it and has permanent magnet, the same way power station generator has many coils with large numbers of turns in each coil. These coils can be used for producing magnetic field just like permanent magnet. There is one stationary coil that stays at a point whereas the other coil is rotational that spins at steady rate with the help of mechanical force that helps to keep it revolving and to produce current and voltage. These mechanical forces can be provided by water, steam and fuel like diesel, coal and gas which helps to rotate the coil through magnetic stationary coil. Power station generators produce very large current and high voltage.

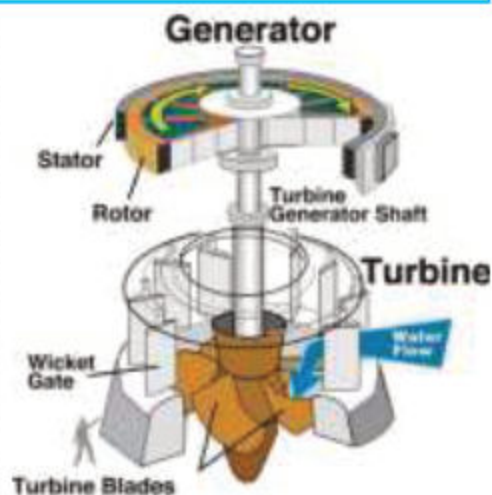


Fig.11.7:Power Station Generator

OTHER SOURCES OF ELECTRICITY

- ✓ List types of energy being used in power stations.

There are many sources other than mechanical generators to produce electricity. Some sources that are being most often used for producing electricity are solar energy, wind energy, nuclear energy and biomass, moving water energy and heat energy.

➤ Solar Energy

Solar panels contain solar cells which are called photo-voltaic cells. The solar powered calculator uses these cells for absorbing sunlight energy and converting into electrical energy to charge the battery or to use it directly. Good thing about solar panels is that they do not pollute the air and environment. In



Fig.11.8: Solar Energy Power Plant

Pakistan, a solar power plant has been installed and named as Quaid-e-Azam solar power plant. One of the draw backs of solar power is that it produces little electricity when sun is behind the clouds and it does not produce any electricity during night time. However, these can be used to recharge or stock electricity in batteries to have an un-interrupted power supply to our homes during night hours. Today, solar panel installations in schools are becoming more popular as it decreases the environmental impact of these institutions and making them greener and more eco-friendly.

➤ Wind Energy

Wind energy is a form of energy which can be used to generate electricity by wind or simply by air pressure. Wind produces the kinetic energy which is used to rotate the blades of fan of wind

energy system. The axil of fan therefore rotates generator to produce electricity. These wind energy systems can be seen on lands which is windy enough to rotate the fan or blades of wind energy system. You can see hurdreds of wind energy power stations around Nooriabad area while travelling from Hyderabad to Karachi.



Fig.11.9: Wind Fan Energy Power Plant

➤ Nuclear energy

Nuclear energy is a form of energy which can be found in the nucleus of an atom of an element. When nuclei of heavy elements are broken into lighter nuclei by a special process called 'fission' a large amount of heat energy is released. This heat makes steam and is used to rotate the turbinewhich runs the electric generator to produce electricity.

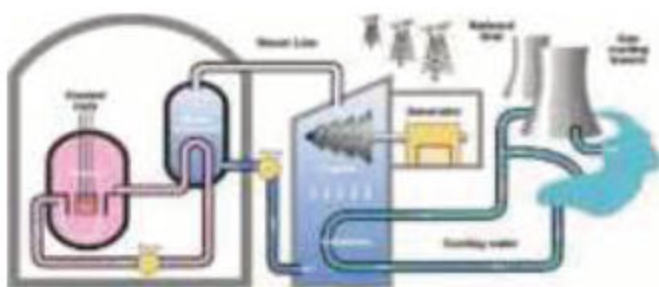


Fig.11.10: Nuclear Energy Plant

➤ Biomass

Biomass is the oldest sources of energy which is obtained from

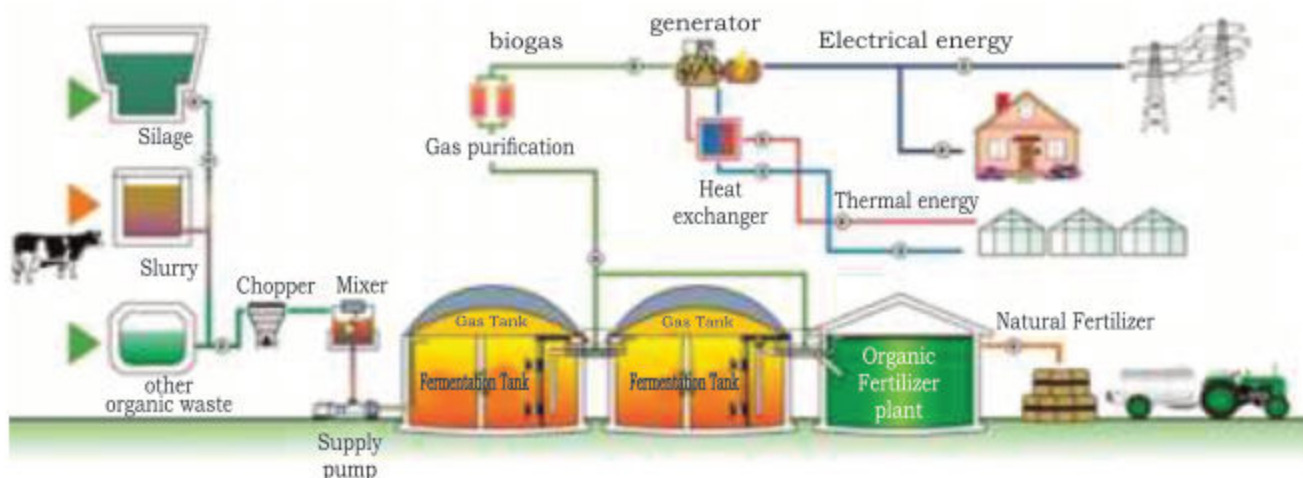


Fig.11.11: Bio-Mass Energy Plant

animal waste and fossils plant material. It is used as a fuel to produce heat energy. The steam then produces kinetic energy in the turbine of biomass and rotate the generators to produce electricity. It is economical way to produce electricity by using biomass.

➤ **Hydro-electricity Energy**

Hydroelectricity is the kind of electricity generated by power of falling or flowing water. The most common type of hydroelectric power plant uses a dam on a river to store water in a reservoir. Water released from the reservoir flows through a turbine, which converts the kinetic energy of falling water into mechanical energy. Then a generator converts the mechanical energy from the turbine into electrical energy.

➤ **Thermal Energy**

Thermal energy is produced from heat. It uses the gas, wood, coal and petroleum to produce heat. This heat is used to generate a pressure in combustion chamber of generator from which a turbine is operated and moves to generate the electricity.

PROBLEMS OF GENERATING ELECTRICITY

✓ Related problems involved in generating electricity.

There are many problems involved in generation of electricity. Some of these are given below:

1. Electricity can be produced by using flow of water. Its running cost is though very low, but the construction of dams is very expensive. Also, the water causes logging in nearby areas and makes the land weak for the agricultural use. Further, in case if there is no raining, there can be water shortage which ultimately causes problem in generating electricity.
2. The nuclear energy is also the source of producing large amount of energy; but it can be very dangerous because of the use of radioactive elements producing radiations and hence, can harm the lives for the years.

3. Thermal energy is a good source of energy; but we need fuel to make heat energy that can be used to run the turbine. The prices of fuels are going very high and that's why this method is very expensive to produce electricity. Also, these fuels when burn, make smoke and hence, pollute the atmosphere.
4. The sources which produce renewable energy are also expensive to install. Their plants like solar energy and wind energy plants are much costly as they produce little amount of electricity.
5. When power is transmitted for homes and commercial areas, the magnetic field around transmission lines may cause hazards for the human health.
6. High voltage shock is harmful. Accidental contact with high light voltage lines can cause harm or death to human and other animals.

INTRODUCTION TO ELECTRONIC SYSTEMS

- ✓Describe basic component of an electronic system.
- ✓Explain A.C and D.C current
- ✓List components that would be needed to turn A.C. to D.C.

Electronics is a branch of physics that deals with controlling of electrical energy. The fundamental role in electronics is of electrons. In our daily life we use lots of electronic devices like, Television, Mobile Phone, Computer, Camera, Radio, Video games, DVD player etc. In this modern era, we have a very advanced electronics appliances that make our life easy and luxurious. Now-a-day we have a package of all necessary things in a single set of mobile phone, for example, a telephone, a camera, a watch, a television, a computer and many more applications.



Fig.11.12: Various Electronic Devices

Alternating Current (AC)

Alternating current is defined as the flow of charge that changes direction periodically. The voltage level also reverses along with the current. Basically AC is used to deliver high power to industries, buildings, etc..

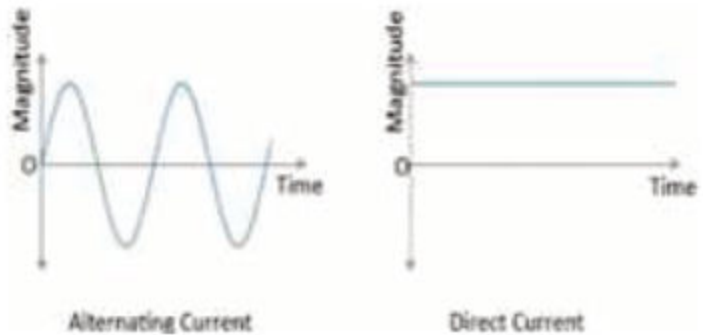


Fig.11.13: Resistor

Direct Current (DC)

The current, which flows in one direction is called Direct current (DC). DC is referred to voltage whose polarity never reverses. It has two conductors with polarity of positive and negative charges.

Conversion of AC to DC (Rectification)

A rectifier is a simple diode or group of diodes which converts alternating current (AC) to direct current (DC). The process is known as rectification. We know that a diode allows electric current in one direction and blocks electric current in another direction. This principle is used in rectification. A half wave rectifier is a type of rectifier which allows only half cycle of the input AC signal while the half cycle is blocked. In this way it converts AC to pulsating DC.

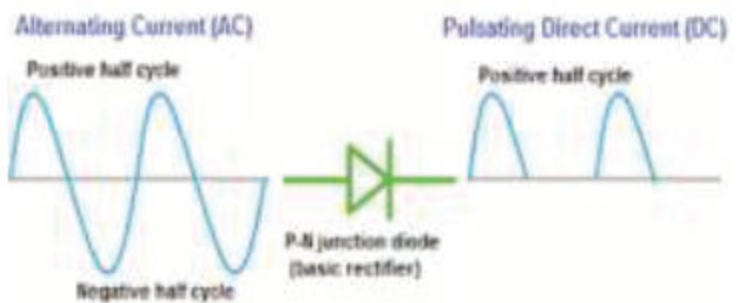


Fig.11.14: Resistor

Basic Components of an Electronic System

The basic components that are used in electronics are Resistor, Diode, Transistor, integrated circuit (IC). These are most commonly used components and are the basic building blocks for electronic circuits.

1. Resistor

Resistor is a device that opposes the flow of electrons in electronic circuits. It is a two terminal device that can be used to reduce the electric current in electronic circuits. The unit of resistance is Ohm and its symbol is (Ω).



Fig.11.15: Resistor

2. Diode

It is made up of a material that is called semiconductor. Semiconductor is the element which acts sometime as conductor and sometime as insulator depending on certain conditions. Diode can be used for different applications like electronics switch, reverse current protector and Alternate current to Direct current conversion. It has different types like LED (Light Emitting Diode) is a diode which gives out the light rays or light beam when an electric current flow through it. This is the same LED which we are using now-a-days at home for lightings to save our electricity. As compared to normal bulb, LED consumes lesser electricity and therefore saves the energy. They are also used in displaying numbers and letters in digital equipment such as calculators, digital clocks, digital meters and many others.



Fig.11.16: Diode

3. Transistor

In electronics transistor is the most important device. A transistor is the device which controls the flow of electricity or simply electrons by acting as a switch that can be turn on and off electronically. It can also be very small that a human eye cannot see. Millions of tiny transistors can be combined into a small device which is called integrated circuit. This can make system more compact and small in size and therefore can be used very fast.

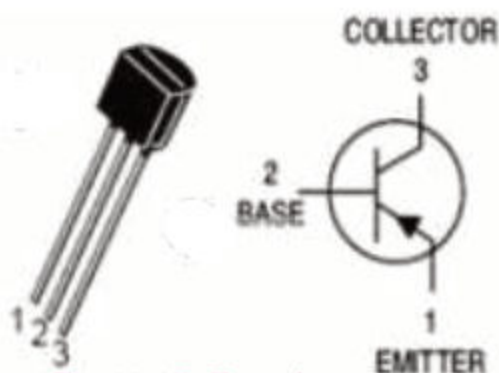


Fig.11.17: Transistors

USES OF COMPONENTS(INPUT, PROCESSOR, OUTPUT)

- ✓ State how output component in various devices could be used in the schools and surroundings

Electronic devices commonly used in our daily life are of three major types according to our requirements; these are given as follow.

1. Input Devices

A device or component that provides information in different forms of energy, converts it to electrical signal and then forwards it to the computer is called input device. There are many types of input devices that can be used for providing input or instruction or information to the other device, for example, the digital camera which takes photo of physical object and converts it to the electrical signal same with the scanner. Keyboard and mouse of a computer are also the input devices which can be used for providing information or the instruction to the computer so that a computer can perform a particular task. Microphone is another example of input device that can be used to convert the voice into electrical signal.



Fig.11.18: Input Devices

2. Processor

It is a component that functions as the brain of a computer or electronic system. In computer, Central Processing Unit (CPU) contains the Processor which performs typical tasks of processing of a computer program and carrying out computer operations. It basically converts the input to the desired output in some calculated way. Your smart phone also has processor which provides the output on screen while having some input in numeric form or through touch screen.



Fig.11.19: Computer Containing Processor

3. Output Devices

An output device is a device that is used to receive the data from processor and shows its results of giving input and processing through processor. For example, when we give input through microphone it converts our voice into electrical signal and acts as processor. Finally, we listen our voice amplified through the speaker which is an output device. Your computer has a monitor; which is also an output device. Smart phone screen and television screen are output devices too. Also, computer monitor, printer, loudspeaker, electric bell and telephone receiver are well-known output devices commonly used in schools, homes and offices.



Fig. 11.20: Output Devices

SUMMARY

1. Electricity is a form of energy produced when electric charges flow through a conductor.
2. Electricity can be generated by moving a magnet through a coil or a coil through a magnet.
3. Power Plant generators are used to generate large amount of electricity to fulfil the domestic and commercial needs of a country.
4. There are some sources which most often use for producing electricity at low cost like, solar energy, wind energy, nuclear energy and biomass energy systems.
5. Electronics is a branch of physics that deals with controlling of electrical energy. The fundamental role in electronics is of electrons.
6. Resistor, Diode, Transistor, integrated circuit (IC) are most commonly used components of electronic circuits.
7. Diode is a device that passes the electric current in one direction and blocks current flow in reverse direction.
8. A transistor is the device which controls the flow of electricity or simply electrons by acting as a switch that can be turned on and off electronically.
9. A device or component that provides information in different forms of energy, converts it to electrical signal and forwards it to computer is called input device.
10. An output device is a device that is used to receive the data from processor and shows its results.

EXERCISE

Q.1 Choose the correct answer:

- i. Which part of a bicycle dynamo spins to generate electricity?
(a) Tyre. (b) magnet. (c) coil. (d) galvanometer.
- ii. The electricity from a cell is called:
(a) Direct current. (b) indirect current.
(c) Alternating current. (d) high current.
- iii. The electricity we use in our homes and schools is called:
(a) Low current. (b) indirect current.
(c) Alternating current. (d) direct current.
- iv. Which of the following is NOT a renewable energy source?
(a) Windpower. (b) geothermal energy.
(c) tidal power. (d) natural gas.
- v. The electronic device which is used as a switch or amplifier is called a:
(a) Transistor. (b) thermostat.
(c) thermistor. (d) transformer.
- vi. IC stands for:
(a) Inner chip. (b) integrated chip.
(c) integrated circuit. (d) input circuit.
- vii. Silicon is an example of:
(a) Insulator. (b) conductor.
(c) semi-conductor. (d) semi-insulator.
- viii. Barcode reader is an:
(a) Output device. (b) input device.
(c) amplifier. (d) insulator.
- ix. Flow of electrons is called:
(a) Electrodes. (b) resistance.
(c) electrolyte. (d) electric current.
- x. When animals and plants are decayed in absence of air, there produces a gas called :
(a) oxygen. (b) bio gas.
(c) carbon dioxide. (d) methane.

Q.2 The stages involved in producing electricity at a power station are given below in a jumbled form. Write the letters of the stages in correct sequence in the provided space below.

(A) The steam turns the turbines.

(B) The electricity is generated and fed to the National Grid.

(C) Fuel is burnt and heats the tanks of water.

(D) The turbines turn the generators.

(E) The water changes to high pressure steam.

Q.3 Generation of electricity by wind turbines is increasing day by day.

Complete the box below by naming the main energy changes in a wind turbine.

Energy in the wind
Energy in the moving rotor blades
Output energy from the turbine

Q.4 Why biofuels are considered as economical sources of energy?

Q.5 What are the basic building blocks for electronic circuits?

Q.6 What are the problems involved in producing electricity by:

(i) Nuclear power generator (ii) Thermal power generator

Q.7 What are the input and output devices? Give some examples

Q.8 Differentiate between Alternating Current and Direct Current.

Q.9 What are the advantages of a Solar Power Plant?

PROJECT

Making a Simple Transformer

Materials needed:

About 1.5 m of thin single core insulated wire; a long iron nail or bolt; a 1.5 V cell; sticky tape; 2.5 V bulb in a holder; small screwdriver

Steps:

1. Cut the wire into two pieces (approximately 0.5 m and 1.0 m in length).
2. Bare the both ends of the wires.
3. Wind the shorter of the two wires around the nail fifty turns, leaving the bared ends free (see diagram below). This is your primary coil.
4. Attach one end of the primary coil to the cell with some sticky tape.
5. Now wrap the second wire around the nail (over the top of the first wire) fifty turns. This is the secondary coil.
6. Attach the two ends of the secondary coil to the bulb holder.
7. Touch (on and off rapidly) the remaining free wire of the primary coil to the free terminal of the cell. What happens?
8. Try again but use seventy-five turns in the secondary coil.
9. Now try other combinations of turns.

